Hyunsik Jeon

AI & Backend Developer Portfolio

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FEATURED PROJECT 01

Top Award — IBM & RedHat AI Transformation Program

Vaccine Daily (AI News Pipeline)

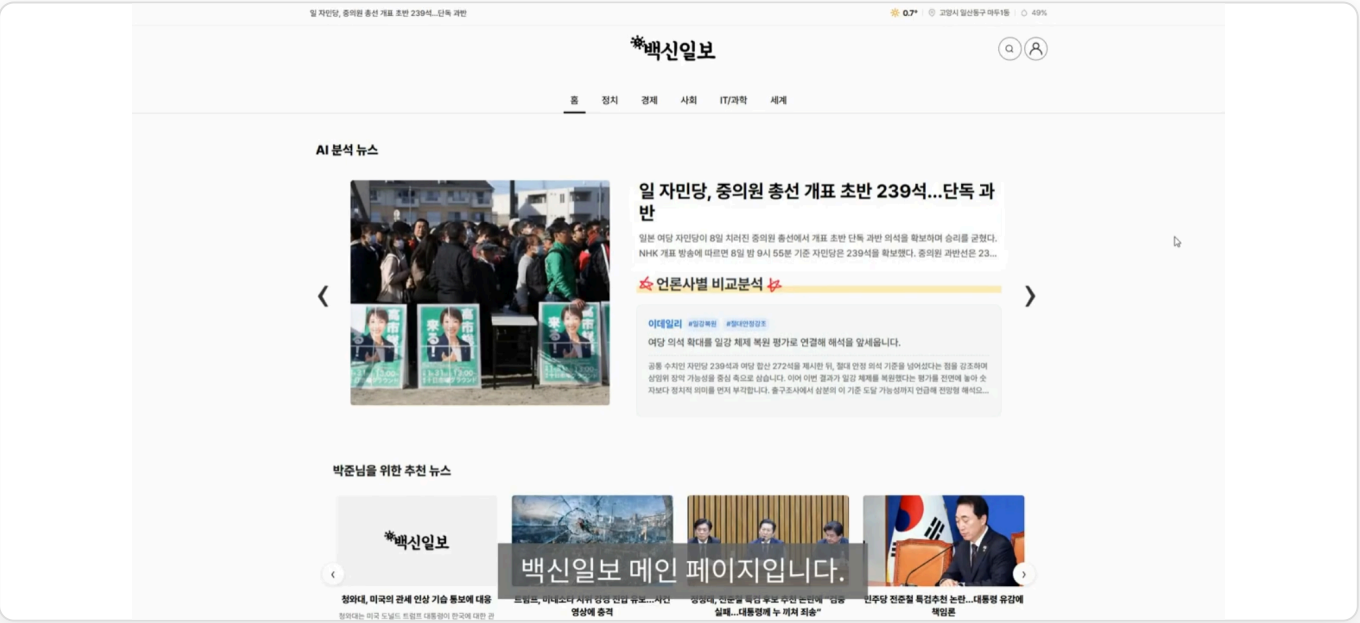
Hyunsik Jeon and 2 others

React · FastAPI · PostgreSQL · ChromaDB · GPT-4o-mini · ko-sroberta · HDBSCAN · GraphRAG · Docker

An automated news intelligence pipeline that crawls 10 major media outlets every 5 minutes and uses a multi-agent workflow to generate verified, hallucination-free personalized reports.

Technical Pipeline Detail

1. **Crawling** Real-time automated data collection from 10 media outlets via Selenium & BS4.
2. **Clustering** Token-efficient group analysis of similar news using HDBSCAN and ko-sroberta.
3. **AI News Article Generator** Multi-Agent AI (Writer → Critic → Editor) for factual verification and summarization.
4. **GraphRAG** Entity-Relation extraction and Knowledge Graph construction for Media Tone of Voice analysis.



FEATURED PROJECT 02

Football AI Prediction System

Hyunsik Jeon (Personal Use)

Python · TensorFlow · Reinforcement Learning · Monte Carlo · XGBoost · LightGBM · FastAPI · Pandas

A high-precision forecasting system analyzing data from 13 European football leagues. It maximizes predictive accuracy by ensembling deep learning models with reinforcement learning strategies.

6-Step System Architecture

Step 1 Data Scrapping: Automated pipeline for match data and global bookmaker odds collection.

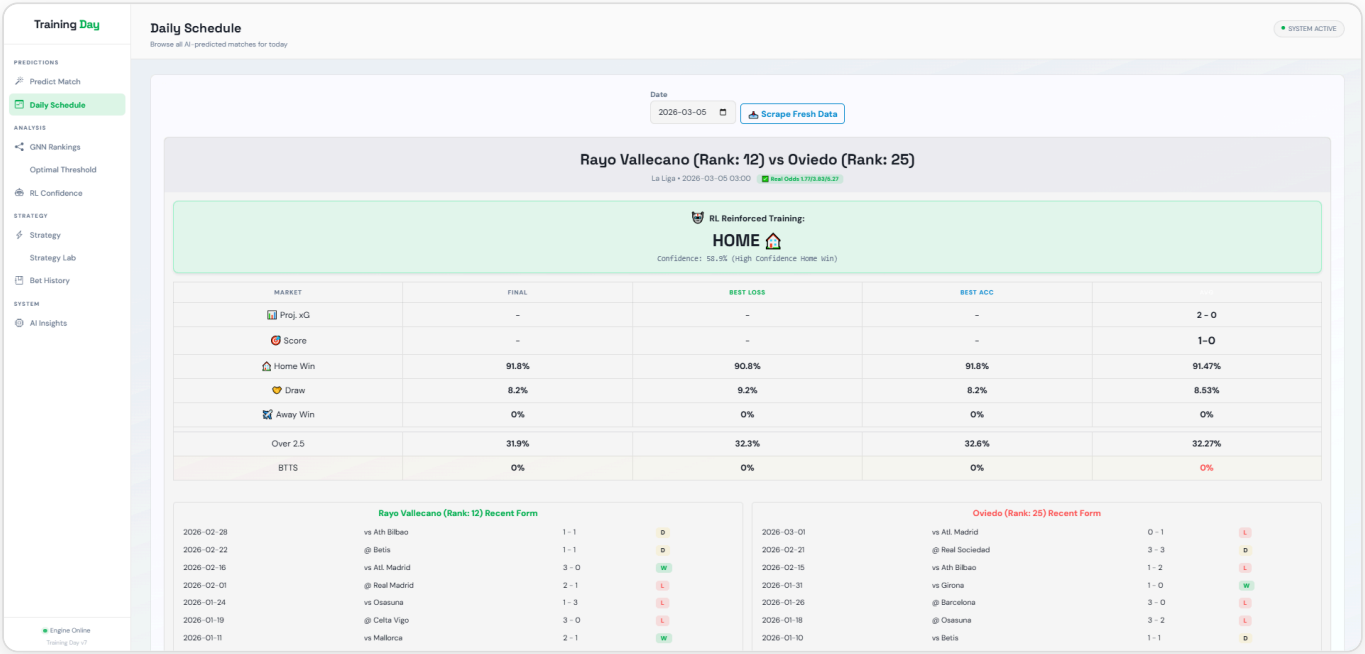
Step 2 Predictive Ensembling: Triple architecture — TensorFlow DL + XGBoost + LightGBM.

Step 3 Confidence Optimization: Reinforcement Learning (RL) agent for tuning logical confidence thresholds.

Step 4 Probabilistic Reasoning: Monte Carlo simulation for calibrated outcome probabilities.

Step 5 Dynamic Tuning: Adaptive sequence length optimization based on league-specific volatility.

Step 6 Power Ranking: ELO-based strength metrics and GNN-based dominance analysis.



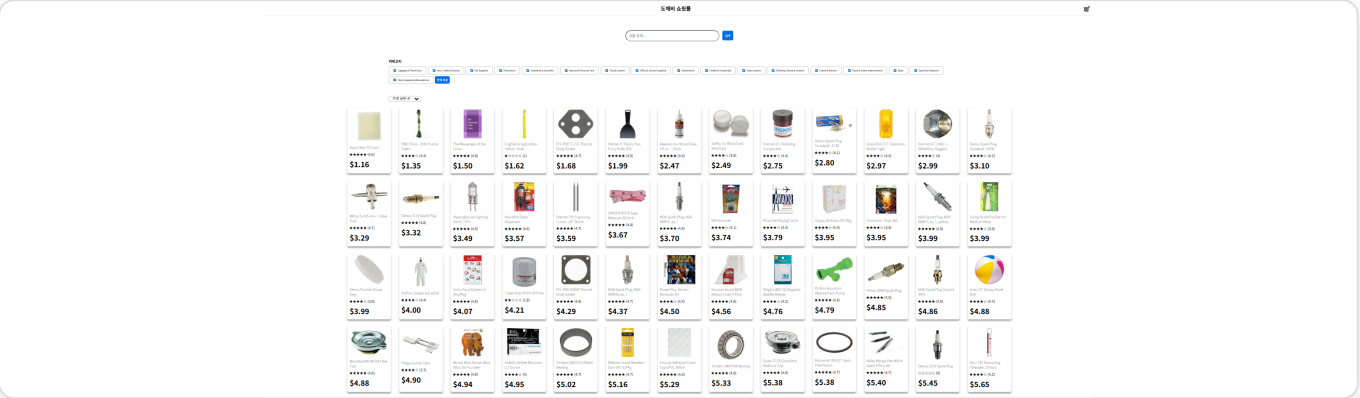
OTHER PROJECTS

Goblin Shopping Mall (E-commerce SPA)

Hyunsik Jeon and 3 others

React · JavaScript · CSS · Vercel

A fully-featured e-commerce platform with 18+ category filters and real-time search. Focused on intuitive UX and high-performance frontend state management.



Crazy Arcade Web Replica

Hyunsik Jeon and 3 others

React · FastAPI · WebSocket · MySQL

A real-time multiplayer game synchronization project using WebSockets. Implemented complex physics and room-state management for smooth gameplay.

My Diary (Emotion Log App)

(Personal Use for Education)

React · Context API · localStorage

A personal journal application focusing on frontend state persistence and custom UI interactions using browser's local storage.